

- (a) Show that if $\mathbf{A}$ has all distinct real eigenvalues, then $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$ is stable if and only if all eigenvalues are nonpositive.
- (b) Show that if $\mathbf{A}$ has all distinct real eigenvalues, then $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$ is asymptotically stable if and only if all eigenvalues are negative.
- (c) Argue that in parts (a) and (b), we can replace “has distinct real eigenvalues” by “is symmetric” and the statements are still true.
50. In an ice tray, the water level in any particular ice cube cell will change at a rate proportional to the *difference* between that cell’s water level and the level in the adjacent cells.

- (a) Argue that a reasonable differentiable equation model for the water levels x , y , and z in the simplified three-cell tray depicted in Figure 9.4 is given by
- $$x' = y - x, \quad y' = x + z - 2y, \quad z' = y - z.$$
- (b) Use eigenvectors to solve this system for the initial conditions $x(0) = 3$, $y(0) = z(0) = 0$.

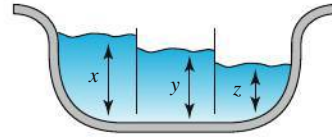


Figure 9.4 Ice tray

9.6 Complex Eigenvalues

In the previous section, we showed that the homogeneous system

$$(1) \quad \mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t),$$

where $\mathbf{A}$ is a constant $n \times n$ matrix, has a solution of the form $\mathbf{x}(t) = e^{rt}\mathbf{u}$ if and only if r is an eigenvalue of $\mathbf{A}$ and $\mathbf{u}$ is a corresponding eigenvector. In this section we show how to obtain two real vector solutions to system (1) when $\mathbf{A}$ is real and has a pair[†] of complex conjugate eigenvalues $\alpha + i\beta$ and $\alpha - i\beta$.

Suppose $r_1 = \alpha + i\beta$ (α and β real numbers) is an eigenvalue of $\mathbf{A}$ with corresponding eigenvector $\mathbf{z} = \mathbf{a} + i\mathbf{b}$, where $\mathbf{a}$ and $\mathbf{b}$ are real constant vectors. We first observe that the complex conjugate of $\mathbf{z}$, namely $\bar{\mathbf{z}} := \mathbf{a} - i\mathbf{b}$, is an eigenvector associated with the eigenvalue $r_2 = \alpha - i\beta$. To see this, note that taking the complex conjugate of $(\mathbf{A} - r_1\mathbf{I})\mathbf{z} = \mathbf{0}$ yields $(\mathbf{A} - \bar{r}_1\mathbf{I})\bar{\mathbf{z}} = \mathbf{0}$ because the conjugate of the product is the product of the conjugates and $\mathbf{A}$ and $\mathbf{I}$ have real entries ($\bar{\mathbf{A}} = \mathbf{A}$, $\bar{\mathbf{I}} = \mathbf{I}$). Since $r_2 = \bar{r}_1$, we see that $\bar{\mathbf{z}}$ is an eigenvector associated with r_2 . Therefore, two linearly independent complex vector solutions to (1) are

$$(2) \quad \mathbf{w}_1(t) = e^{r_1 t} \mathbf{z} = e^{(\alpha + i\beta)t} (\mathbf{a} + i\mathbf{b}),$$

$$(3) \quad \mathbf{w}_2(t) = e^{r_2 t} \bar{\mathbf{z}} = e^{(\alpha - i\beta)t} (\mathbf{a} - i\mathbf{b}).$$

As in Section 4.3, where we handled complex roots to the auxiliary equation, let's use one of these complex solutions and Euler's formula to obtain two real vector solutions. With the aid of Euler's formula, we rewrite $\mathbf{w}_1(t)$ as

$$\begin{aligned} \mathbf{w}_1(t) &= e^{\alpha t} (\cos \beta t + i \sin \beta t) (\mathbf{a} + i\mathbf{b}) \\ &= e^{\alpha t} \{ (\cos \beta t \mathbf{a} - \sin \beta t \mathbf{b}) + i (\sin \beta t \mathbf{a} + \cos \beta t \mathbf{b}) \}. \end{aligned}$$

We have thereby expressed $\mathbf{w}_1(t)$ in the form $\mathbf{w}_1(t) = \mathbf{x}_1(t) + i\mathbf{x}_2(t)$, where $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ are the two *real* vector functions

$$(4) \quad \mathbf{x}_1(t) := e^{\alpha t} \cos \beta t \mathbf{a} - e^{\alpha t} \sin \beta t \mathbf{b},$$

$$(5) \quad \mathbf{x}_2(t) := e^{\alpha t} \sin \beta t \mathbf{a} + e^{\alpha t} \cos \beta t \mathbf{b}.$$

[†]Recall that the complex roots of a polynomial equation with real coefficients must occur in complex conjugate pairs.

Since $\mathbf{w}_1(t)$ is a solution to (1), then

$$\begin{aligned}\mathbf{w}'_1(t) &= \mathbf{A}\mathbf{w}_1(t), \\ \mathbf{x}'_1 + i\mathbf{x}'_2 &= \mathbf{A}\mathbf{x}_1 + i\mathbf{A}\mathbf{x}_2.\end{aligned}$$

Equating the real and imaginary parts yields

$$\mathbf{x}'_1(t) = \mathbf{A}\mathbf{x}_1(t) \quad \text{and} \quad \mathbf{x}'_2(t) = \mathbf{A}\mathbf{x}_2(t).$$

Hence, $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ are real vector solutions to (1) associated with the complex conjugate eigenvalues $\alpha \pm i\beta$. Because $\mathbf{a}$ and $\mathbf{b}$ are not both the zero vector, it can be shown that $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ are linearly independent vector functions on $(-\infty, \infty)$ (see Problem 15).

Let's summarize our findings.

Complex Eigenvalues

If the real matrix $\mathbf{A}$ has complex conjugate eigenvalues $\alpha \pm i\beta$ with corresponding eigenvectors $\mathbf{a} \pm i\mathbf{b}$, then two linearly independent real vector solutions to $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$ are

$$(6) \quad e^{\alpha t} \cos \beta t \mathbf{a} - e^{\alpha t} \sin \beta t \mathbf{b},$$

$$(7) \quad e^{\alpha t} \sin \beta t \mathbf{a} + e^{\alpha t} \cos \beta t \mathbf{b}.$$

Example 1 Find a general solution of

$$(8) \quad \mathbf{x}'(t) = \begin{bmatrix} -1 & 2 \\ -1 & -3 \end{bmatrix} \mathbf{x}(t).$$

Solution The characteristic equation for $\mathbf{A}$ is

$$|\mathbf{A} - r\mathbf{I}| = \begin{vmatrix} -1-r & 2 \\ -1 & -3-r \end{vmatrix} = r^2 + 4r + 5 = 0.$$

Hence, $\mathbf{A}$ has eigenvalues $r = -2 \pm i$.

To find a general solution, we need only find an eigenvector associated with the eigenvalue $r = -2 + i$. Substituting $r = -2 + i$ into $(\mathbf{A} - r\mathbf{I})\mathbf{z} = \mathbf{0}$ gives

$$\begin{bmatrix} 1-i & 2 \\ -1 & -1-i \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The solutions can be expressed as $z_1 = 2s$ and $z_2 = (-1+i)s$, with s arbitrary. Hence, the eigenvectors associated with $r = -2 + i$ are $\mathbf{z} = s \operatorname{col}(2, -1+i)$. Taking $s = 1$ gives the eigenvector

$$\mathbf{z} = \begin{bmatrix} 2 \\ -1+i \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} + i \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

We have found that $\alpha = -2$, $\beta = 1$, and $\mathbf{z} = \mathbf{a} + i\mathbf{b}$ with $\mathbf{a} = \operatorname{col}(2, -1)$, and $\mathbf{b} = \operatorname{col}(0, 1)$, so a general solution to (8) is

$$\begin{aligned}\mathbf{x}(t) &= c_1 \left\{ e^{-2t} \cos t \begin{bmatrix} 2 \\ -1 \end{bmatrix} - e^{-2t} \sin t \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} \\ &\quad + c_2 \left\{ e^{-2t} \sin t \begin{bmatrix} 2 \\ -1 \end{bmatrix} + e^{-2t} \cos t \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} \\ (9) \quad \mathbf{x}(t) &= c_1 \begin{bmatrix} 2e^{-2t} \cos t \\ -e^{-2t}(\cos t + \sin t) \end{bmatrix} + c_2 \begin{bmatrix} 2e^{-2t} \sin t \\ e^{-2t}(\cos t - \sin t) \end{bmatrix}. \quad \blacklozenge\end{aligned}$$

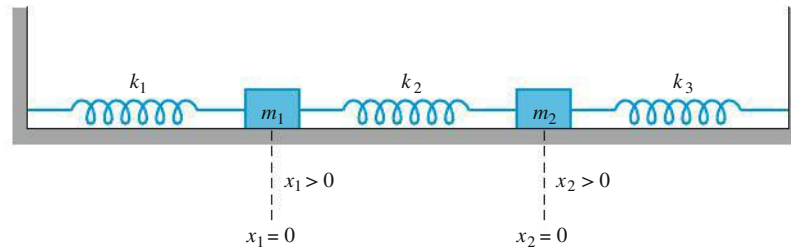


Figure 9.5 Coupled mass-spring system with fixed ends

Complex eigenvalues occur in modeling coupled mass-spring systems. For example, the motion of the mass-spring system illustrated in Figure 9.5 is governed by the second-order system

$$(10) \quad \begin{aligned} m_1 x_1'' &= -k_1 x_1 + k_2(x_2 - x_1), \\ m_2 x_2'' &= -k_2(x_2 - x_1) - k_3 x_2, \end{aligned}$$

where x_1 and x_2 represent the displacements of the masses m_1 and m_2 to the right of their equilibrium positions and k_1, k_2, k_3 are the spring constants of the three springs (see the discussion in Section 5.6). If we introduce the new variables $y_1 := x_1, y_2 := x_1', y_3 := x_2, y_4 := x_2'$, then we can rewrite the system in the normal form

$$(11) \quad \mathbf{y}'(t) = \mathbf{A}\mathbf{y}(t) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -(k_1 + k_2)/m_1 & 0 & k_2/m_1 & 0 \\ 0 & 0 & 0 & 1 \\ k_2/m_2 & 0 & -(k_2 + k_3)/m_2 & 0 \end{bmatrix} \mathbf{y}(t).$$

For such a system, it turns out that $\mathbf{A}$ has only imaginary eigenvalues and they occur in complex conjugate pairs: $\pm i\beta_1, \pm i\beta_2$. Hence, any solution will consist of sums of sine and cosine functions. The frequencies of these functions

$$f_1 := \frac{\beta_1}{2\pi} \quad \text{and} \quad f_2 := \frac{\beta_2}{2\pi}$$

are called the **normal** or **natural frequencies** of the system (β_1 and β_2 are the **angular frequencies** of the system).

In some engineering applications, the only information that is required about a particular device is a knowledge of its normal frequencies; one must ensure that they are far from the frequencies that occur naturally in the device's operating environment (so that no resonances will be excited).

Example 2 Determine the normal frequencies for the coupled mass-spring system governed by system (11) when $m_1 = m_2 = 1$ kg, $k_1 = 1$ N/m, $k_2 = 2$ N/m, and $k_3 = 3$ N/m.

Solution To find the eigenvalues of $\mathbf{A}$, we must solve the characteristic equation

$$|\mathbf{A} - r\mathbf{I}| = \begin{vmatrix} -r & 1 & 0 & 0 \\ -3 & -r & 2 & 0 \\ 0 & 0 & -r & 1 \\ 2 & 0 & -5 & -r \end{vmatrix} = r^4 + 8r^2 + 11 = 0.$$

From the quadratic formula we find $r^2 = -4 \pm \sqrt{5}$, so the four eigenvalues of $\mathbf{A}$ are $\pm i\sqrt{4 - \sqrt{5}}$ and $\pm i\sqrt{4 + \sqrt{5}}$. Hence, the two normal frequencies for this system are

$$\frac{\sqrt{4 - \sqrt{5}}}{2\pi} \approx 0.211 \quad \text{and} \quad \frac{\sqrt{4 + \sqrt{5}}}{2\pi} \approx 0.397 \text{ cycles per second. } \blacklozenge$$

9.6 EXERCISES

In Problems 1–4, find a general solution of the system $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$ for the given matrix $\mathbf{A}$.

1. $\mathbf{A} = \begin{bmatrix} 2 & -4 \\ 2 & -2 \end{bmatrix}$ 2. $\mathbf{A} = \begin{bmatrix} -2 & -5 \\ 1 & 2 \end{bmatrix}$

3. $\mathbf{A} = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$

4. $\mathbf{A} = \begin{bmatrix} 5 & -5 & -5 \\ -1 & 4 & 2 \\ 3 & -5 & -3 \end{bmatrix}$

In Problems 5–9, find a fundamental matrix for the system $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$ for the given matrix $\mathbf{A}$.

5. $\mathbf{A} = \begin{bmatrix} -1 & -2 \\ 8 & -1 \end{bmatrix}$ 6. $\mathbf{A} = \begin{bmatrix} -2 & -2 \\ 4 & 2 \end{bmatrix}$

7. $\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$

8. $\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -13 & 4 \end{bmatrix}$

9. $\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$



In Problems 10–12, use a linear algebra software package to compute the required eigenvalues and eigenvectors for the given matrix $\mathbf{A}$ and then give a fundamental matrix for the system $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$.

10. $\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 13 & -4 & -12 & 4 \end{bmatrix}$

11. $\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -2 & 2 & -3 & 2 \end{bmatrix}$

12. $\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -29 & -4 \end{bmatrix}$

In Problems 13 and 14, find the solution to the given system that satisfies the given initial condition.

13. $\mathbf{x}'(t) = \begin{bmatrix} -3 & -1 \\ 2 & -1 \end{bmatrix} \mathbf{x}(t),$

(a) $\mathbf{x}(0) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ (b) $\mathbf{x}(\pi) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

(c) $\mathbf{x}(-2\pi) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ (d) $\mathbf{x}(\pi/2) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

14. $\mathbf{x}'(t) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix} \mathbf{x}(t),$

(a) $\mathbf{x}(0) = \begin{bmatrix} -2 \\ 2 \\ -1 \end{bmatrix}$ (b) $\mathbf{x}(-\pi) = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

15. Show that $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ given by equations (4) and (5) are linearly independent on $(-\infty, \infty)$, provided $\beta \neq 0$ and $\mathbf{a}$ and $\mathbf{b}$ are not both the zero vector.

16. Show that $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ given by equations (4) and (5) can be obtained as linear combinations of $\mathbf{w}_1(t)$ and $\mathbf{w}_2(t)$ given by equations (2) and (3). [Hint: Show that

$$\mathbf{x}_1(t) = \frac{\mathbf{w}_1(t) + \mathbf{w}_2(t)}{2}, \quad \mathbf{x}_2(t) = \frac{\mathbf{w}_1(t) - \mathbf{w}_2(t)}{2i}.$$

In Problems 17 and 18, use the results of Problem 42 in Exercises 9.5 to find a general solution to the given Cauchy–Euler system for $t > 0$.

17. $t\mathbf{x}'(t) = \begin{bmatrix} -1 & -1 & 0 \\ 2 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix} \mathbf{x}(t)$

18. $t\mathbf{x}'(t) = \begin{bmatrix} -1 & -1 \\ 9 & -1 \end{bmatrix} \mathbf{x}(t)$

19. For the coupled mass–spring system governed by system (10), assume $m_1 = m_2 = 1$ kg, $k_1 = k_2 = 2$ N/m, and $k_3 = 3$ N/m. Determine the normal frequencies for this coupled mass–spring system.
20. For the coupled mass–spring system governed by system (10), assume $m_1 = m_2 = 1$ kg, $k_1 = k_2 = k_3 = 1$ N/m, and assume initially that $x_1(0) = 0$ m, $x'_1(0) = 0$ m/sec, $x_2(0) = 2$ m, and $x'_2(0) = 0$ m/sec. Using matrix algebra techniques, solve this initial value problem.
21. **RLC Network.** The currents in the RLC network given by the schematic diagram in Figure 9.6 are governed by the following equations:

$$\begin{aligned} 4I'_2(t) + 52q_1(t) &= 10, \\ 13I_3(t) + 52q_1(t) &= 10, \\ I_1(t) &= I_2(t) + I_3(t), \end{aligned}$$

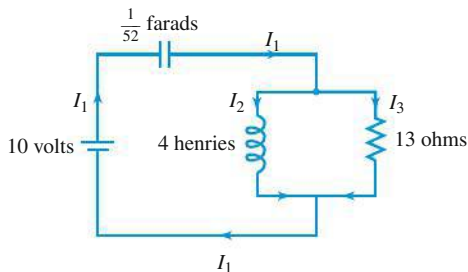


Figure 9.6 RLC network for Problem 21

where $q_1(t)$ is the charge on the capacitor, $I_1(t) = q'_1(t)$, and initially $q_1(0) = 0$ coulombs and $I_1(0) = 0$ amps. Solve for the currents I_1 , I_2 , and I_3 . [Hint: Differentiate the first two equations, eliminate I_1 , and form a normal system with $x_1 = I_2$, $x_2 = I'_2$, and $x_3 = I_3$.]

22. **RLC Network.** The currents in the RLC network given by the schematic diagram in Figure 9.7 are governed by the following equations:

$$\begin{aligned} 50I'_1(t) + 80I_2(t) &= 160, \\ 50I'_1(t) + 800q_3(t) &= 160, \\ I_1(t) &= I_2(t) + I_3(t), \end{aligned}$$

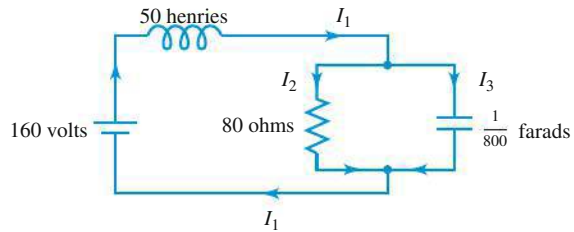


Figure 9.7 RLC network for Problem 22

where $q_3(t)$ is the charge on the capacitor, $I_3(t) = q'_3(t)$, and initially $q_3(0) = 0.5$ coulombs and $I_3(0) = 0$ amps. Solve for the currents I_1 , I_2 , and I_3 . [Hint: Differentiate the first two equations, use the third equation to eliminate I_3 , and form a normal system with $x_1 = I_1$, $x_2 = I'_1$, and $x_3 = I_2$.]

23. **Stability.** In Problem 49 of Exercises 9.5, (page 542), we discussed the notion of stability and asymptotic stability for a linear system of the form $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$. Assume that $\mathbf{A}$ has all distinct eigenvalues (real or complex).
- (a) Show that the system is stable if and only if all the eigenvalues of $\mathbf{A}$ have nonpositive real part.
- (b) Show that the system is asymptotically stable if and only if all the eigenvalues of $\mathbf{A}$ have negative real part.
24. (a) For Example 1, page 535, verify that
- $$\mathbf{x}(t) = c_1 \begin{bmatrix} -e^{-2t} \cos t + e^{-2t} \sin t \\ e^{-2t} \cos t \end{bmatrix} + c_2 \begin{bmatrix} -e^{-2t} \sin t - e^{-2t} \cos t \\ e^{-2t} \sin t \end{bmatrix}$$
- is another general solution to equation (8).
- (b) How can the general solution of part (a) be directly obtained from the general solution derived in (9) on page 535?

9.7 Nonhomogeneous Linear Systems

The techniques discussed in Chapters 4 and 6 for finding a particular solution to the nonhomogeneous equation $y'' + p(x)y' + q(x)y = g(x)$ have natural extensions to nonhomogeneous linear systems.

Undetermined Coefficients

The method of undetermined coefficients can be used to find a particular solution to the nonhomogeneous linear system

$$\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{f}(t)$$